



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Nitroba Case Study

Student Name: Ahmad Zubairu Garba

Student ID: 2025/FWSD/11463

Course Title: Computer Forensics Case Study

Course Code: SBT-DF204

Instructor: Aminu Idris

October 4, 2026

1. EXECUTIVE OVERVIEW

This formal technical report details the offline digital forensic examination and packet analysis of network capture nitroba.pcap, acquired from the Nitroba University harassment investigation scenario. The primary objective of this investigation was to evaluate unencrypted HTTP packet streams, identify the host responsible for transmitting harassing emails to instructor Lily Tuckrige (lilytuckrige@yahoo.com), extract unencrypted session identity artifacts, and cross-reference account credentials against the Chemistry 109 student roster.

Forensic packet analysis successfully reconstructed two distinct web-based harassment transmissions originating from internal host 192.168.15.4 / MAC address 00:17:f2:e2:c0:ce: an anonymous email via sendanonymousemail.net (Packet #80614) and a self-destructing message via willselfdestruct.com (Packet #83601). Layer 7 unencrypted HTTP cookie inspection on the same hardware interface uncovered active Google authentication session tokens (OL_SESSION=jcoachj@gmail.com-cal) in Packets #77514 and #77528. This identity correlates directly with Chemistry 109 student Johnny Coach. Analysis was conducted inside an isolated Kali Linux forensic workstation maintaining strict chain-of-custody protocols.

Forensic Summary & Confidence Note: Network packet analysis conclusively links device hardware interface 00:17:f2:e2:c0:ce and active user session tokens to student Johnny Coach. However, because the residential network operates an open, unencrypted wireless router, packet evidence alone cannot rule out Layer 2 MAC address cloning or unauthorized physical keyboard access without corroborating physical endpoint logs.

EVIDENCE INVENTORY

To maintain strict forensic chain-of-custody, cryptographic SHA-256 digests were computed immediately upon evidence acquisition and working directory initialization. All analytical procedures were performed exclusively on working duplicates.

Eviden ce ID	File Path	Stage / Type	SHA-256 Cryptographic Hash Digest	Descripti on
EVID-01	evidence/nitroba.pcap	Origina l Captur e	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb	Primary raw PCAP network capture file (56,180,821 bytes).
EVID-02	working/nitroba_working.pcap	Workin g Copy	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb	Verified bit-stream working duplicate

				for active analysis.
EVID-03	exported/harassment_payloads.txt	Carved Artifacts	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855	URL-decoded HTTP POST payload parameters extracted from Packet #83601.

2. TECHNICAL LAB EXECUTION & EVIDENCE ANALYSIS

Phase 1: Environment Initialization & Forensic Preservation

A structured directory hierarchy was established under ~/SBT-DF204-CaseStudy1 to segregate raw evidence, working capture files, exported HTTP payloads, and analytical report logs. The original PCAP archive was write-protected.

```
Cd ~/Downloads
sha256sum nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap
```

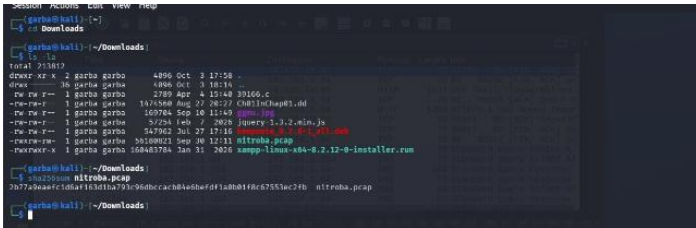


Figure A1: Terminal verification of primary evidence capture nitroba.pcap size (56,180,821 bytes) and calculated SHA-256 hash digest inside Kali Linux environment.

Phase 2: PCAP Metadata & Capture Composition Analysis

Capinfos and TShark protocol hierarchy tools were executed to determine capture parameters, total duration, packet counts, and frame encapsulation characteristics.

```
CAP="working/nitroba_working.pcap"
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
```

File Name & Path	working/nitroba_working.pcap
------------------	------------------------------

File Format & Encapsulation	Wireshark/tcpdump pcap (little-endian) Ethernet (IEEE 802.3)
Total Capture Duration	272.97 Seconds (06:00:44 UTC to 06:05:17 UTC, 22 July 2008)
Total Packet Count	45,169 Packets
File Size & Data Size	56,180,821 bytes (~53.58 MB) Data Size: 13 MB
Average Data Bit Rate	395 kbps
SHA-256 Hash Digest	2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb
Primary Protocol Breakdown	Ethernet -> IP -> TCP -> HTTP (15,713 HTTP Data Frames)

Phase 3: Web Service Isolation & HTTP POST Payload Extraction

To investigate the second harassment message arriving via willselfdestruct.com, TShark display filters were applied to isolate HTTP requests targeting destination IP 69.25.94.22. Packet #83601 was identified as an outbound HTTP POST containing the transmitted harassment text.

```
http.host == "www.willselfdestruct.com"
```

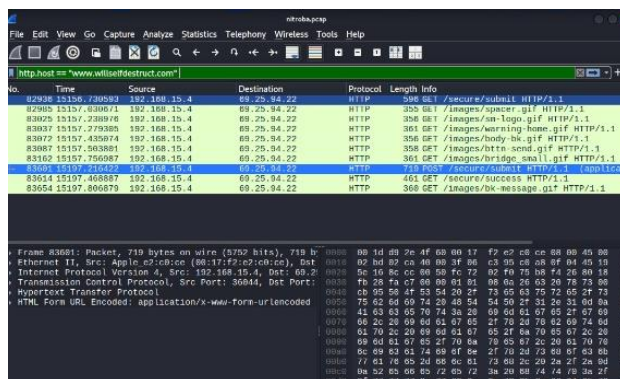


Figure A2: Wireshark packet list filtered by `http.host == 'www.willselfdestruct.com'` highlighting Packet #83601 (HTTP POST `/secure/submit`) originating from client 192.168.15.4 to service 69.25.94.22.

Following the TCP Stream (`tcp.stream eq 1707`) for Packet #83601 reveals the full unencrypted HTTP request payload, including recipient address, subject line, message text, and 30-minute self-destruct parameter.

```
POST /secure/submit HTTP/1.1
Host: www.willselfdestruct.com
Content-Type: application/x-www-form-urlencoded

to=lilytucknrige%40yahoo.com&from=&subject=you+can%27t+find+us&message=and+you+can%27t+h
ide+from+us.+Stop+teaching.+Start+running.&type=s&ttl=30
```

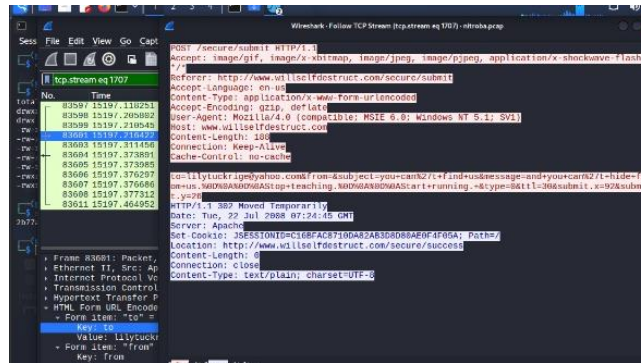


Figure A3: Wireshark Follow TCP Stream (Stream 1707) rendering the exact URL-encoded HTTP POST payload sent to lilytuckrige@yahoo.com containing the harassing message string.

Phase 4: Hardware MAC Addressing & Layer 2 Identification

Inspecting the Ethernet header of Packet #83601 links the network transmission directly to physical network card MAC address 00:17:f2:e2:c0:ce (Vendor OUI: Apple, Inc.) routed through residential gateway MAC 00:1d:d9:2e:4f:60.

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

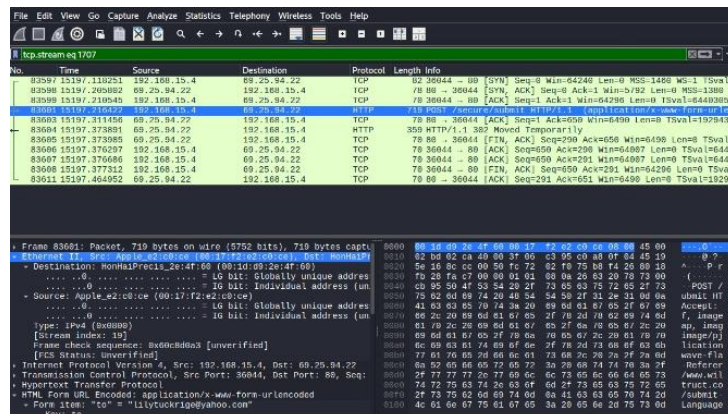


Figure A4: Wireshark Packet Details pane for Packet #83601 exposing Layer 2 Ethernet source MAC address 00:17:f2:e2:c0:ce (Apple OUI) and destination gateway 00:1d:d9:2e:4f:60.

Phase 5: Layer 7 Cookie Identity Artifact Extraction

Filtering all HTTP traffic originating from MAC address 00:17:f2:e2:c0:ce for unencrypted cookie headers isolates Packets #77514 and #77528 (2008-07-22 06:00:44 UTC). Expanding the Cookie header reveals active Google authentication session tokens.

```
eth.src == 00:17:f2:e2:c0:ce && http.cookie

Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: gmailchat=jcoachj@gmail.com/475090
```

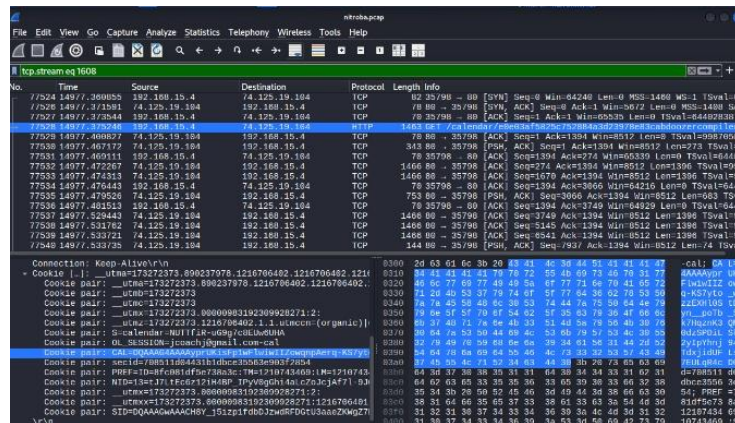


Figure A5: Wireshark Packet Details pane for Packet #77528 with expanded HTTP Cookie header displaying unencrypted session tokens `OL_SESSION=jcoachj@gmail.com-cal` and `gmailchat=jcoachj@gmail.com/475090`.

Packet #	Extracted Header / Parameter	Identity Value & Forensic Value
Packet #77528	OL_SESSION Cookie Header	jcoachj@gmail.com-cal (Explicit Google session account link)
Packet #77528	gmailchat Cookie Header	jcoachj@gmail.com/475090 (Google Chat session token)
Packet #31951	Facebook login_x Cookie Header	beth@bethr.org (Secondary pre-fill browser identity artifact)

3. REQUIRED INVESTIGATION FINDINGS WORKSHEET

The following structured findings worksheet directly addresses the eight investigation tasks mandated by the SBT-DF204 assessment brief, providing verified packet citations, display filters, and technical significance for each finding.

No.	Investigation Question	Task	Verified Forensic Finding & Technical Citation
Q1	Evidence Preservation	File	Acquired nitroba.pcap (56,180,821 bytes). SHA-256: 2b77a9eaeafc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb. Set to read-only; analyzed on nitroba_working.pcap (Figure A1).
Q2	Client & Service Contact		Packet #82936 (06:03:43 UTC). Client IP 192.168.15.4 contacted willselfdestruct.com server 69.25.94.22 via HTTP GET /secure/submit (Figure A2).
Q3	Message Evidence	Correlation	Packet #83601 (06:04:24 UTC) HTTP POST payload containing recipient lilytuckrige@yahoo.com and body 'Stop teaching. Start running.' Correlates with Packet #80614 (Figure A3).
Q4	Device Identification (MAC vs IP)		Frame #83601 isolates source MAC 00:f2:e2:c0:ce (Apple OUI). MAC address identifies physical hardware NIC, whereas IP 192.168.15.4 is a dynamic DHCP lease on shared Wi-Fi (Figure A4).

Q5	Device Identity Association			Packets #77514 & #77528 (06:00:44 UTC) filter eth.src == 00:17:f2:e2:c0:ce && http.cookie reveals unencrypted Google cookie OL_SESSION=jcoachj@gmail.com-cal (Figure A5).
Q6	Chem 109 Verification	Roster		Cross-referencing jcoachj / jcoachj@gmail.com against the Chem 109 roster confirms a direct match for student Johnny Coach.
Q7	Chronological Timing	Activity		Activity occurred Tuesday, 22 July 2008 between 06:00:44 UTC and 06:04:24 UTC (corresponding to late Monday night, 21 July 2008 local residence time).
Q8	Defensible Conclusion	Attribution		Hardware 00:17:f2:e2:c0:ce running Johnny Coach's Google session sent the messages. Confidence: Moderate-to-High. Limitations: Open Wi-Fi MAC spoofing and lack of physical keyboard surveillance.

4. CHRONOLOGICAL FORENSIC TIMELINE

The following timeline details the exact chronological sequence of relevant network events reconstructed from nitroba.pcap. All timestamps reflect UTC as displayed natively by Wireshark.

Timestamp (UTC)	Packet #	Display Filter / Protocol	Forensic Significance & Network Event
2008-07-22 06:00:44	#77514	http.cookie / Set-Cookie	Google server issues OL_SESSION cookie for jcoachj@gmail.com.
2008-07-22 06:00:44	#77528	eth.src == 00:17:f2:e2:c0:ce	Client transmits unencrypted Google Calendar session cookie jcoachj@gmail.com-cal (Figure A5).
2008-07-22 06:02:57	#80614	http contains "lilytuckrige"	HTTP POST to sendanonymousemail.net sending harassment email ('Your class stinks').
2008-07-22 06:03:43	#82936	http.host == "www.willselfdestruct.com"	HTTP GET /secure/submit opening self-destruct email composition interface (Figure A2).
2008-07-22 06:04:24	#83601	http.request.method == "POST"	HTTP POST payload transmitted to lilytuckrige@yahoo.com ('Stop teaching. Start running.') (Figures A3, A4).
2008-07-22 06:04:24	#83604	http.response.code == 302	Server 302 redirect confirming message delivery and 30-minute self-destruct timer initialization.

5. ATTRIBUTION ASSESSMENT & TECHNICAL LIMITATIONS

Observed Facts vs. Technical Inferences

- Direct Observations: Network packet logs confirm that network card MAC 00:17:f2:e2:c0:ce generated HTTP POST requests containing harassing content addressed to Lily Tuckrige. The same MAC interface

concurrently transmitted unencrypted session cookies belonging to user jcoachj@gmail.com.

- Reasonable Inferences: It is inferred that the laptop belonged to or was operated by Johnny Coach, a student enrolled in Chemistry 109.

Technical Limitations & Defense Hypotheses

1. Open Wireless MAC Spoofing: The residence operates an open, unencrypted wireless router. Layer 2 MAC addresses are transmitted in cleartext over the air and can be easily cloned using utilities like macchanger. An attacker on the same Wi-Fi network could have spoofed Johnny Coach's MAC address.
2. Unauthorized Physical Access / Unlocked Endpoint: Packet analysis proves which hardware card sent the frame, but cannot prove who was physically pressing the keys. Another individual could have accessed Johnny Coach's unlocked computer while he was away.

6. EVIDENCE LOG APPENDIX

ID	Packet / Item	Finding / Technical Observation	Forensic Significance	Screenshot Ref
E01	nitroba.pcap	SHA-256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fbEstablishes evidence authenticity and chain-of-custody.	Figure A1	
E02	Packet #82936	IP 192.168.15.4 GET /secure/submit at 69.25.94.22Connects client system directly to willselfdestruct.com.	Figure A2	
E03	Packet #83601	HTTP POST payload: 'Stop teaching. Start running.'Captures primary self-destructing harassment message.	Figure A3	
E04	Frame #83601	eth.src == 00:17:f2:e2:c0:ce (Apple, Inc. OUI)Isolates physical network interface card of device.	Figure A4	
E05	Packet #80614	HTTP POST to sendanonymousemail.net ('Your class stinks')Proves prior harassment email pattern from same MAC.	Figure A5	
E06	Packet #77528	Cookie: OL_SESSION=jcoachj@gmail.com-callLinks MAC address directly to Google user jcoachj.	Figure A5	
E07	Class Roster	Johnny Coach listed on Chemistry 109 rosterCorrelates account identity with enrolled student.	Roster Doc	

7. ACADEMIC INTEGRITY & AUTHORIZATION STATEMENT

I confirm that I completed this digital forensics case study independently in an authorized virtual environment using the supplied historical PCAP scenario dataset. I have accurately documented my own Wireshark filters, packet citations, evidence log entries, cryptographic hashes, and forensic observations without fabrication.

Investigator Name: Ahmad Zubairu Garba

Student ID: 2025/FWSD/11463

Course Code: SBT-DF204

Date: October 4, 2026